consume, customized market data products that suit their particular needs. This flexibility addresses concerns that consolidated market data is overly broad or unnecessarily complex because it allows competing consolidators and their subscribers to adjust the breadth and complexity of the market data products they offer and consume, respectively.[132] On the other hand, limiting consolidated market data to only depth of book data, certain odd-lot information, and auction data, as one commenter suggested, would not include regulatory data—such as information regarding trading halts and price bands—that the Commission believes is necessary to trade effectively and efficiently.[133]

In response to comments recommending a more streamlined or flexible process to include additional data elements in core data,[134] the Commission agrees that the definition of consolidated market data should permit additional data elements to be added pursuant to effective national market system plan amendments. However, the Commission continues to believe that this process should be limited to future regulatory, administrative, or self-regulatory organization-specific program information.[135] As discussed below,[136] the transaction and quotation information reflected in the definition of core data—including best bids and offers, the NBBO, protected quotations, last sale data, depth of book data, and auction information—is

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132 See infra Section III. See also infra notes 139 and 140 and accompanying text (discussing the Commission's adoption of the new defined term "consolidated market data product").

133 See infra Section II.H.

134 See Clearpool Letter at 11.

135 See Proposing Release, 85 FR at 16734.

136 See infra Sections II.C through II.G.

specified in the rule.[137] The rule as proposed and adopted is designed to account appropriately for additional regulatory, administrative, and self-regulatory organization-specific program information data elements that may emerge periodically through the approval of new SRO rules or the development and refinement of technical specifications to be included in consolidated market data through the effective national market system plan amendment process.[138]

The Commission is defining a new term, "consolidated market data product" to mean any data product developed by a competing consolidator that contains consolidated market data or components of consolidated market data. The definition of consolidated market data product also specifies that components of consolidated market data include the enumerated elements, and any subcomponent of the elements, of consolidated market data in §242.600(b)(19) and that all consolidated market data products must reflect data consolidated across all national securities exchanges and national securities associations.[139] As discussed further below, Rule 614 will require competing consolidators to offer one or more consolidated market data products to their subscribers, and will not, as proposed, require them to offer a product that contains all elements of consolidated market data.[140] In addition, the Commission recognizes that some market

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137 The Commission will continue to monitor the usefulness of these core data elements to market participants and consider whether any modifications to the definition of core data are necessary or appropriate as the markets evolve. Interested persons also may petition the Commission to amend such definition if they believe particular changes are warranted.

138 See infra Sections II.H, II.J, and II.K. Both SRO rule changes and effective national market system plan amendments are subject to the public notice and comment process, as well as Commission review. See Exchange Act Section 19(b)(1), 15 U.S.C. 78s(b)(1); Rule 19b-4, 17 CFR 240.19b-4; Rule 608(b) of Regulation NMS, 17 CFR 242.608(b).

139 See infra Section VIII.

140 See infra Section III.C.8(a).

participants will not want or need a consolidated market data product that contains all elements of consolidated market data.

3. The Fifth Amendment's Takings Clause

The Constitution's Takings Clause prevents the taking of private property for public use without just compensation.[141] One commenter stated that the proposal to expand the data that would be required to be provided under Regulation NMS would violate the Takings Clause by "effecting a physical taking … without just compensation."[142] The commenter asserted that the proposal would require it to "turn over vast amounts of their proprietary market data—valuable property that Nasdaq currently sells to market participants at a reasonable rate of return—to competing consolidators and self-aggregators at prices set by the operating committee of the consolidated NMS plan."[143] The commenter stated that the government would "expropriate property belonging to Nasdaq and redistribute it to Nasdaq's competitors at prices set, in part, by the non-SRO members of the consolidated NMS plan's operating committee" that would be "laboring under a conflict-of-interest and would have no incentive to pay 'just compensation' for the property taken from Nasdaq."[144]

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141 U.S. Const. amend. 5 ("No person shall be held to answer for a capital, or otherwise infamous crime, unless on a presentment or indictment of a grand jury, except in cases arising in the land or naval forces, or in the militia, when in actual service in time of war or public danger; nor shall any person be subject for the same offense to be twice put in jeopardy of life or limb; nor shall be compelled in any criminal case to be a witness against himself, nor be deprived of life, liberty, or property, without due process of law; nor shall private property be taken for public use, without just compensation.").

142 Nasdaq Letter IV at 50.

143 Id.

144 Id. at 50–51.

Neither the expansion of NMS information pursuant to the definition of consolidated market data nor the requirement that national securities exchanges and associations make the data necessary to generate consolidated market data available to competing consolidators and self-aggregators constitutes a taking for the following reasons. The Commission's action does not encroach on or appropriate any property. The exchanges developed their proprietary data within a highly regulated statutory and regulatory structure that provides the Commission with ample authority to decide—and revise—which types of information the exchanges must provide to market participants to fulfill their responsibilities under the Exchange Act.[145] Moreover, the SROs will be compensated for making the data necessary to generate consolidated market data available to competing consolidators and self-aggregators pursuant to fees established by the effective national market system plan(s). Even if non-SRO members of plan Operating Committees have a degree of authority to influence proposed consolidated market data fees, the Commission retains authority to ensure that those fees are "fair and reasonable" and "not unreasonably discriminatory." The exchanges thus had no reasonable basis to expect that the current regulatory structure would remain in place in perpetuity in this highly regulated field, and, in any event, they will not be deprived of the economic benefits of the information they will provide to market participants.[146]

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145 See supra note 5.

146 See Ruckleshaus v. Monsanto Co., 467 U.S. 986, 1005–07 (1984) (noting that the reasonableness of an investment-backed expectation depends in part on whether the regulated activity has been in an area "that has long been the source of public concern and the subject of government regulation"); District Intown Properties Ltd. P'ship v. District of Columbia, 198 F.3d 874, 884 (D.C. Cir. 1999) ("Businesses that operate in an industry with a history of regulation have no reasonable expectation that regulation will not be strengthened to achieve established legislative ends."); Me. Educ. Ass'n Benefits

C. Definition of "Core Data" under Rule 600(b)(21)

1. Proposal

As stated in the Proposing Release,[147] Regulation NMS does not contain a definition of "core data," although various Regulation NMS rules describe the information that is required to be collected, consolidated, and disseminated under Regulation NMS.[148] The Commission proposed defining "core data" to include the information currently referred to as core data—last sale data, each SRO's best bid and best offer ("BBO"), and the NBBO[149]—along with new information that is not currently required to be provided under Regulation NMS or by the exclusive SIPs. The proposed new information included quotation data for smaller-sized orders in higher-priced stocks (pursuant to a new definition of "round lot"), information on certain quotations below the best bid or above the best offer (pursuant to a new definition of "depth of book data"), and information about orders participating in auctions (pursuant to a new definition of "auction information"). Specifically, the proposed definition of core data included: (A) quotation sizes; (B) aggregate quotation sizes; (C) best bid and best offer; (D) national best bid and national best offer; (E) protected bid and protected offer; (F) transaction reports; (G) last sale data; (H) odd-lot transaction data disseminated pursuant to the effective national market system plan or plans required under §242.603(b) as of [date of Commission approval of this Adopting Release]; (I) depth of book data; and (J) auction information.

Tr. v. Cioppa, 695 F.3d 145, 154 (1st Cir. 2012) (the plaintiff's "expectations are substantially diminished by the highly regulated nature of the industry in which it operates").

147 See Proposing Release, 85 FR at 16730.

148 See, e.g., Rules 601, 602, and 603 of Regulation NMS.

149 See supra note 16.

Additionally, the proposed definition of core data specified how odd-lots are to be aggregated for purposes of certain data elements included within the definition of core data. Specifically, the proposed definition stated that the best bid and best offer, national best bid and national best offer, and depth of book data shall include odd-lots that when aggregated are equal to or greater than a round lot, and that such aggregation shall occur across multiple prices and shall be disseminated at the least aggressive price of all such aggregated odd-lots.[150]

Finally, the proposed definition of core data did not include certain information—specifically, OTC Bulletin Board ("OTCBB") data, and corporate bond and index data—that is currently provided by the exclusive SIPs.[151]

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150 As discussed below, the proposed definition of core data also specified an odd-lot aggregation methodology for protected quotations. See infra Section II.E.2(b).

151 See Proposing Release, 85 FR at 16736.

2. Final Rule and Response to Comments

(a) Expansion of Core Data, Generally

Multiple commenters supported the expansion of NMS information generally[152] and of core data[153] in particular.[154] One commenter, "agree[ing] that the proposed information to be included in core data has become much more important to broker-dealers in recent years…," "strongly support[ed] expanding core data to include additional information of significance to investors."[155] Another commenter stated that "add[ing] more pricing information to the consolidated tape … would be a fundamental improvement that would expand data access to

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152 See T. Rowe Price Letter at 1–2 ("Expanding the content of NMS information would improve its utility when consumed electronically (e.g., by algorithmic trading systems or smart order routers)."); BlackRock Letter at 2 ("BlackRock is supportive of expanding and revamping the content of NMS information. We agree that this would help to reduce information asymmetries between market participants who rely upon SIP data and those who purchase proprietary data feeds from the national securities exchanges.").

153 See Clearpool Letter at 11 (supporting "the inclusion of this additional information in core data, which can reduce the reliance on exchanges' proprietary data feeds and provide market participants with additional information to make informed order routing and execution decisions," while also "recommend[ing] that the Commission require a 'retail interest indicator' to be added to quotes to assist market participants in defining what portion of the quote is attributable to retail interest"); RBC Letter at 4 (stating that RBC "generally support[s] the Proposal's definition of Core Data"); letter from Tim Lang, Chief Executive Officer, ACS Execution Services, LLC, to Vanessa Countryman, Secretary, Commission, dated May 26, 2020, ("ACS Execution Services Letter") at 2; IEX Letter at 2 ("We support the Market Infrastructure Proposal because it will update the content of 'core data' to better reflect the information needed to participate in today's markets…."); ICI Letter at 4 ("We support the Commission expanding the scope of core data, which will benefit funds and their shareholders.").

154 As discussed below, many commenters also expressed views on the specific elements of the definition of core data. See infra Sections II.D; II.E; II.F; II.G.

155 ACS Execution Services Letter at 2.

Main Street investors in a very meaningful way."[156] A different commenter said that "all data is 'core data.'"[157]

Some commenters opposed the expansion of core data, however. One commenter, though agreeing that Regulation NMS should define core data, stated that the new proposed core data elements are not necessary or useful for all market participants but will raise the costs of core data for all market participants by requiring them to receive and process core data to meet their regulatory obligations.[158] Similarly, another commenter, though supportive of the Commission formally defining core data in its regulations, argued that the proposed definition was "poorly designed" because it "only consider[s] the requirements of market participants that need, and are able to consume, a richer data set" and that the proposed definition "would require non-professional investors who do not need such rich data to purchase and consume even more unnecessary data elements (e.g., depth of book data) than the current SIP product provides."[159] Another commenter argued that the Commission falsely assumed that the decision by some market participants to supplement current core data with proprietary data means that this additional data is necessary to all market participants and investors, and that the expanded set of information included in the proposed definition of core data "is neither necessary nor relevant to the business models and trading or investment strategies of many, if not most, ordinary investors

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156 Letter from Jeffrey T. Brown, Senior Vice President Legislative and Regulatory Affairs, Charles Schwab & Co., Inc., to Vanessa Countryman, Secretary, Commission, dated May 26, 2020, ("Schwab Letter") at 2–3.

157 Virtu Letter at 2, 5 ("[T]he 'core data' offered through the SIPs is no longer sufficient for most market participants to trade competitively in today's marketplace.").

158 See TD Ameritrade Letter at 3.

159 NYSE Letter II at 3–4.

and market participants."[160] Additionally, the commenter stated that the Commission "failed to collect data regarding whether any meaningful number of market participants that desire access to non-core data are actually unable to obtain it, either directly from exchanges or indirectly (and often free of charge) from their brokers."[161]

The Commission is adopting the definition of core data largely as proposed, as discussed further below.[162] In the Proposing Release, the Commission stated its preliminary belief that the content of core data has not kept pace with market developments and that the proposed expansion of core data would enhance its usefulness to address the needs of a broad cross-section of market participants.[163] Comments received from a variety of market participants—including exchanges, buy-side firms, and sell-side firms—have borne this out. Numerous commenters expressed support for the proposed definition of core data, stating that the specific subcomponents of core data, such as five levels of depth of book data, would help market participants to trade more effectively.[164] Several commenters also pointed out that expanding

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160 Nasdaq Letter IV at 7–8.

161 Id. at 8 (footnote removed). See also NYSE Letter II at 3–8; TD Ameritrade Letter at 4.

162 The Commission is revising the proposed definition of core data to include odd-lots priced at or better than the NBBO, to specify how quotation sizes are to be displayed in core data, and to require SRO attribution of core data elements. The Commission is also modifying the proposed definitions of depth of book data and auction information and is not adopting the proposed amendments to the definition of protected bid or protected offer, which definitions are embedded in the definition of core data. The particular elements of the definition of core data are discussed below. See infra Sections II.C.2(b); II.D; II.E; II.F; II.G.

163 See Proposing Release, 85 FR at 16735–76.

164 See, e.g., T. Rowe Price Letter at 2 ("We believe the addition of depth of book data (specifically, the five price levels above the protected offer and below the protected bid) and auction imbalance information, including opening, reopening, and closing auctions, will make SIP data a much more viable alternative to proprietary market data.… This

core data would promote a wider dissemination of this data, including to market participants who cannot afford expensive proprietary feeds.[165] For the reasons discussed in the Proposing Release and as set forth in detail below with respect to the specific elements of core data,[166] the

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additional data will help reduce the information asymmetries that currently exist between SIP data and proprietary data."); ACS Execution Services Letter at 2 ("ACS strongly supports expanding core data to include additional information of significance to investors. As the proposal notes, through the provision of such additional information, market participants may have access to data to make better routing and trading decisions.").

165 See, e.g., Virtu Letter at 5 ("[I]ncluding depth of book information in the SIP will allow investors who cannot afford to pay for costly Exchange proprietary feeds to trade more competitively in the marketplace, and we believe five levels of depth of book is a reasonable and appropriate place to land."); Clearpool Letter at 11 ("[C]urrently, the 'core data' provided through the SIP only includes the NBBO and top-of-book data. For this reason, there continues to be no viable alternatives for broker-dealers to paying exchanges for their proprietary market data, both to provide competitive execution services to clients and, equally important, to meet best execution obligations. Clearpool therefore strongly supports the inclusion of this additional information in core data, which can reduce the reliance on exchanges' proprietary data feeds and provide market participants with additional information to make informed order routing and execution decisions."); IEX Letter at 5–6 ("For these reasons, the NBBO no longer encompasses the 'core data' that market participants need to stay competitive and satisfy best execution responsibilities. The fact that depth of book data can only be obtained through exchange proprietary data feeds allows exchanges to charge extraordinarily high prices completely disproportionate to any reasonable estimation of the cost of producing that data. … Importantly, however, to the extent that a significant subset of market participants could rely on this data as a viable alternative to purchasing proprietary data, or could viably choose to purchase less proprietary data than they need today, it could help to harness market competition to restrain data fee increases that today are largely unrestrained."); letter from James J. Angel, Associate Professor of Finance, Georgetown University, to the Commission, dated June 12, 2020, ("Angel Letter") at 7–8 ("Providing data on a visibly level playing field will increase public trust in the integrity of the markets…. Freely available information about the entire market, including orders inside the spread and the depth of book, will reduce the asymmetry of information in the market between small retail investors and larger players. This added transparency will reduce the notion that markets are 'rigged' in favor of larger players.").

166 See Proposing Release, 85 FR at 16735–59 (discussing market developments such as rising stock prices and increased odd-lot trading, decimalization, and the growth of auctions and the need to expand core data to include smaller-sized orders in higher priced